

PAVINDRAN VELALAGAN

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 [pavindranvelalagan.github.io](https://github.com/pavindranvelalagan) |  [pavindranvelalagan](https://www.linkedin.com/in/pavindranvelalagan) |  [pavindranvelalagan](https://www.youtube.com/pavindranvelalagan)

EDUCATION

- **University of Peradeniya** Jun 2023 - Present
BSc (Hons) in Computer Engineering | Current GPA: 3.31/4.00 Kandy, Sri Lanka
 - Societies: IEEE Computer Society, AeroGenZ Aviation Club, Perabeats Media Club

EXPERIENCE

- **Cogntix**  August 2025 - October 2025
ML Engineering Apprentice (Internship) Hybrid
 - Worked on text extraction techniques for complex, multi-column CV layouts.
 - Analyzed and tested methods for AI-based CV filtering to explore better candidate matching.
 - Contributed to the development of a document classification model to validate user uploads (CV vs. non-CV).
 - Gained hands-on experience with cloud deployment by building and hosting a CV parsing application using AWS (EC2, Route 53, Lightsail, Lambda).

PROJECTS

- **Smart IV - End-to-End Infusion Monitoring & Control System (Group Project)**  Jan 2026 - Present
Rust/Tauri, ESP32, MQTT, C++, Hardware Design, IoT
 - Developed a closed-loop IV system using an IR drop counter and stepper motor mechanism to monitor and automatically regulate flow rate.
 - Integrated IoT connectivity via MQTT with mobile and centralized desktop applications for real-time monitoring and management.
 - Contributed to the Rust/Tauri desktop application and developed firmware for the edge device.
 - **Recognition:** Smart IV received multiple national-level awards, runner up placements, and finalist recognitions.
- **Cross-Resource Interference in Heterogeneous ARM Systems (Research Project)** Sep 2026 - Present
Linux, C/C++, Python, Bash, perf, PMU Counters, GPU Profiling, ARM, Nanopore Genomics
 - Investigating CPU-GPU-memory interference in heterogeneous systems, focusing on performance degradation caused by shared resources such as memory bandwidth, cache, DRAM, interconnects, power, and thermal capacity.
 - Designing controlled co-execution experiments to compare isolated and concurrent CPU, GPU, and memory-intensive workloads and quantify their impact on execution time, throughput, and performance isolation.
 - Applying Linux performance-engineering tools and hardware performance counters to analyze CPU/GPU utilization, IPC, cache misses, memory bandwidth, stalls, memory traffic, and other indicators to identify the underlying causes of workload interference.
 - Exploring interference-aware resource-management techniques such as CPU affinity, workload co-scheduling, concurrency control, cgroups, and resource throttling, with planned validation using realistic Nanopore genomic-processing workloads.
- **Data Analytics Web App**  Apr 2018 - Jun 2018
Python (Django), JavaScript, PHP, MySQL
 - Solo development of a web app allowing CSV/Excel data import and generation of complex graphs.
 - Selected as a finalist in the Yarl Geek Challenge 2018.

PUBLICATIONS

J=JOURNAL

- [J.1] Pavindran Velalagan, et al. (2025). **ViT-RoT: Vision Transformer-Based Robust Framework for Tomato Leaf Disease Recognition**. *MDPI - AgriEngineering, Q1 Journal*. Published Jun 10, 2025.

SKILLS

- Programming Languages: Python, JavaScript, C, SQL
- Web Technologies: React.js, Node.js, HTML/CSS, REST APIs
- Hardware & Embedded: Raspberry Pi, ESP32
- Tools & Cloud: AWS (EC2, Lambda, Lightsail), Git/GitHub, VS Code
- Data Science/ML: OpenCV, NumPy, Google Gemini API

ACHIEVEMENTS

- **Winner - Best Innovative Product (Central Province) at SLASSCOM National Ingenuity Awards 2026** Jun 2026
SLASSCOM (Sri Lanka Association for Software and Services Companies)
 - Awarded for Smart IV, a Medical IoT device developed as our 3rd Year Project.
- **1st Runner Up - NetX IoT Challenge 2026** Sep 2026
University of Sri Jayewardenepura IEEE SB, IEEE ComSoc
 - 1st Runner Up in the IoT competition with the project Smart IV.
- **Finalist - Innov IoT Challenge 2026** Sep 2026
SLIIT
 - Finalist in the IoT competition with the project Smart IV.
- **Finalist - SLIoT Challenge 2026** Feb 2026
University of Moratuwa, SLT-MOBITEL, IESL
 - Finalist in the national IoT competition with the project Smart IV.
- **Finalist - Startup Spark 2024** Oct 2024
Techno 2024, IESL Student Chapter - University of Moratuwa
 - Finalist for the Intelligent Traffic Monitoring System (ITMS) project, proposing an AI-driven solution with dynamic traffic signal control and emergency vehicle prioritization.
- **Finalist - Yarl Geek Challenge (Season 7)** 2018
Yarl IT Hub
 - Selected as a finalist for the Data Analytics Web Application project.

VOLUNTEER EXPERIENCE

- **Student Branch Ambassador** Aug 2026 – Present
IEEE Xtreme 20.0
 - Selected as a global Student Branch Ambassador representing the university for the 20th edition of IEEE Xtreme, the world’s largest 24-hour virtual programming competition.
 - Facilitate team registration, drive community engagement, and coordinate outreach efforts to encourage student participation.
- **Invigilator** Jan 2026
ICPC (International Collegiate Programming Contest) Sri Lanka Peradeniya Regional Contest 2025/26
 - Served as an invigilator for the first-ever ICPC Regional programming contest hosted in Sri Lanka.
- **Casual Instructor** Nov 2024 – Feb 2025
University of Peradeniya
 - Assisted undergraduates in CO1010 Programming for Engineers, covering Python fundamentals and practical problem-solving.
- **Webmaster** Nov 2025 – Present
IEEE CS Student Chapter - UoP
 - Manage and maintain the chapter’s web presence.
 - **Previous Role:** Marketing Team Member (Jul 2024 – Nov 2025). Contributed to graphic design and promotional campaigns for events like AlgoForge.
- **Marketing and Public Relations Coordinator** Jan 2025 – Jan 2026
AerogenZ

CREATIVE EXPERIENCE

- **Founder & Photographer** Sep 2023 - Present
Vivid Dreams
 - Established and operate a freelance photography brand specializing in portraiture and event photography.
- **Videography Team (Videographer / Video Editor / Crew)** Dec 2023 - Jan 2026
Perabeats
 - Contributed as a Video Editor and Videographer (Dec 2023 - Jan 2025) and Crew Member (Jan 2025 - Jan 2026).